Java Data Objects
by
Robin M. Roos

Source Code Archive (JDOSource.JAR) version 1.0.1

INTRODUCTION

This document accompanies the Source Code Archive for the book "Java Data Objects", by Robin M. Roos.

INSTALLATION

Extract the archive to an empty directory of your choice. The scripts provided in the archive expect the directory to be C:\JDOSource. You will need to edit the scripts if you use a different installation directory, although this is not at all difficult.

You can extract the archive using JAR or using WinZip, whichever you prefer to use.

```
mkdir \JDOSource
cd \JDOSource

path below is the directory into which JDOSource.jar was saved

jar xvf path\JDOSource.jar
```

STRUCTURE

Once extracted, the directory structure is as follows:

`C:\JDOSource\bin`	Windows batch files for environment setup, compilation and enhancement
`C:\JDOSource\classes`	Output directory for compilation
`C:\JDOSource\config`	Properties files pertaining to specific JDO Implementations
`C:\JDOSource\db`	Location of the HSQL database
`C:\JDOSource\ddl`	Output directory for .SQL scripts to create database schema
`C:\JDOSource\doc`	This document
`C:\JDOSource\dtd`	Location of the Document Type Descriptor (jdo.dtd)
`C:\JDOSource\enhanced`	Output directory for the enhancement process
`C:\JDOSource\lib`	Location of useful library files

`C:\JDOSource\source`	Source code for the examples on a per-chapter basis

EXAMPLE USAGE

Some readers will find it useful to follow one worked example. Here I provide a step-by-step guide to configuring and running the first example.

Start a command prompt window. Set the PATH so that it begins with the JDOSource\bin directory. This will enable you to run the scripts provided.

```
SET PATH=c:\JDOSource\bin;%PATH%
```

Then run the setup script for the implementation you are using. For this example I am using LiDO. The setup script sets the CLASSPATH to include the LiDO classes.

```
LIDO
```

Then go to the source directory of your choice.

```
cd \JDOSource\Source\Ch02-SimpleExample\0201-SimpleExample
```

Before compiling the classes it is useful to run the jdocleanup script. This removes the compiled and enhanced classes, and deletes the HSQL database.

```
jdocleanup
```

Now compile the source code for the domain and application classes.

```
compile *.java
```

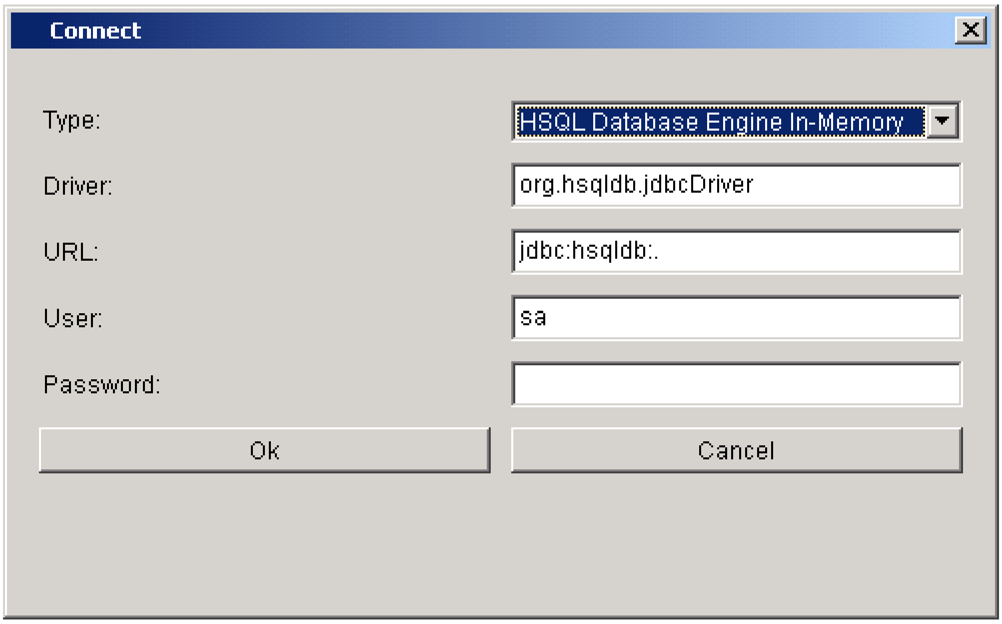
The next step is to enhance the classes. The enhance-lido script actually performs two functions. Firstly it enhances the compiled classes according to the given persistence descriptor (BusinessPartner.jdo) in this instance. Secondly, it analyses the class structure to generate a SQL schema in the form of a .SQL script. The .SQL script is placed into the c:\JDOSource\ddl directory.

```
enhance-lido BusinessPartner.jdo
```

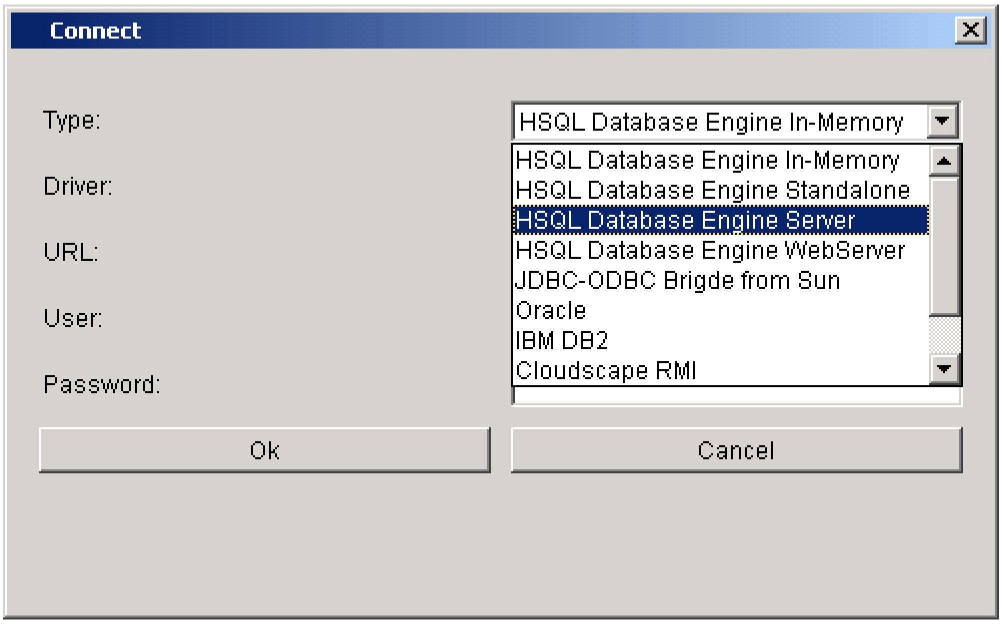
Now it's time to build the database. The cleanup script deleted any previous database, so when the server is started the database will be empty.

You can use any database supported by your implementation. I have chosen to use HSQL here. Start the database server and then start the admin tool.

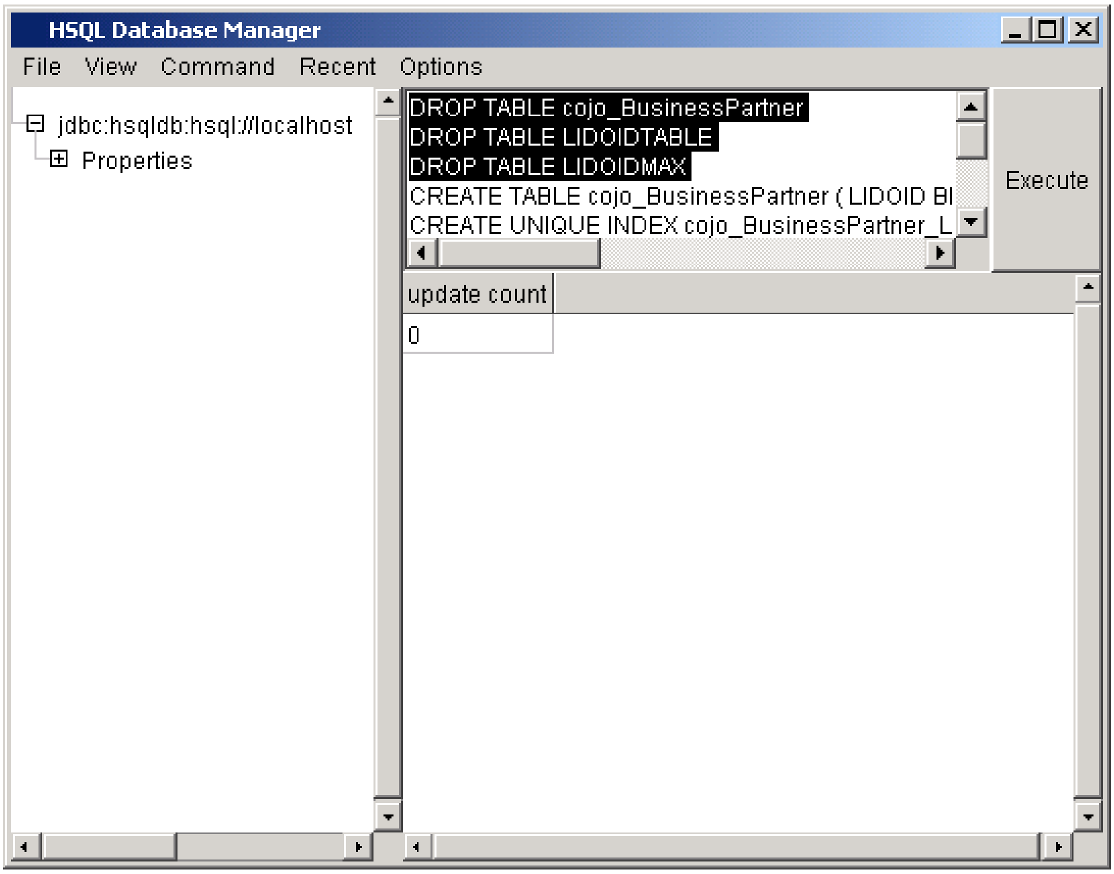
```
Start hsqlserver
Start hsqladmin
```



Once the admin tool is running, connect to the server by selecting Hsql Server in the drop-down combo box.

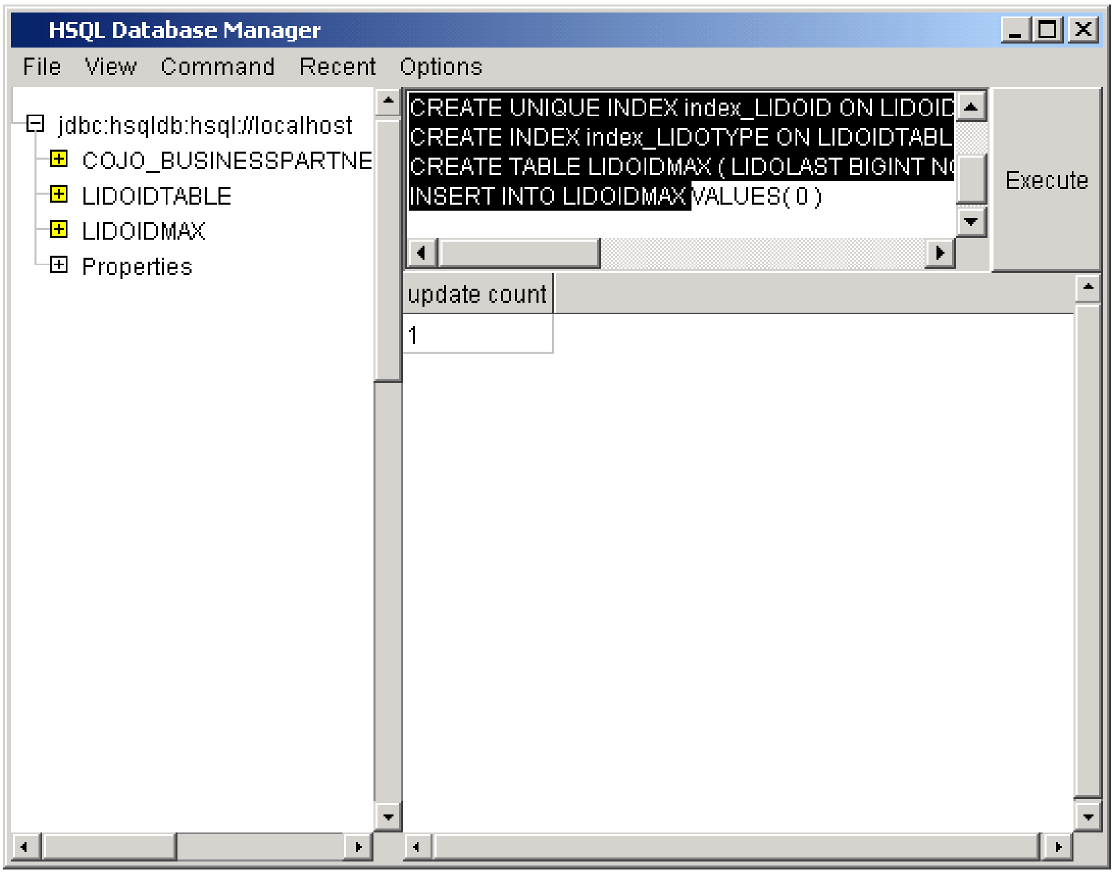


Click OK to connect to the already-running server. Then open the script c:\JDOSource\ddl\lido.sql.



Note that it begins by DROPping tables. Since the database is empty to start with you will need to remove the DROP statements from the command window before the script can be executed.

To actually see the tables that have been created you must select View->Refresh in the menu.



Once the database has been built you can run the application. LiDO requires a property name to be set indicating the name of the persistence descriptor. The LiDO property for this is called jdo.metadata. The TestBusinessPartner application takes three parameters; the partnerNumber, the name and the address. Thus the execution command is:

```
java -Djdo.metadata=BusinessPartner.jdo
      com.ogilviepartners.jdobook.app.TestBusinessPartner
      1 "Robin Roos" "Milton Keynes"
```

The output from executing the application will vary slightly depending on the implementation you are using. Here are my results with LiDO (the significant output is in bold at the end):

```
LiDO C:\JDOSource\source\Ch02-SimpleExample\0201-SimpleExample>java
-Djdo.metadata=BusinessPartner.jdo
com.ogilviepartners.jdobook.app.TestBusinessPartner 1 "Robin Roos"
"Milton Keynes"

Listing standard JDO properties
javax.jdo.PersistenceManagerFactoryClass=
com.libelis.lido.PersistenceManagerFactory
Listing non-standard JDO properties
javax.jdo.option.connectionDriverName=org.hsqldb.jdbcDriver
lido.sql.maxbatch=0
javax.jdo.option.nontransactionalWrite=false
javax.jdo.option.retainValues=true
lido.maxPool=10
javax.jdo.option.connectionUserName=sa
```

```
javax.jdo.option.metadata=metadata.jdo
javax.jdo.option.msWait=5
javax.jdo.option.connectionPassword=
lido.sql.poolsize=10
javax.jdo.option.optimistic=false
javax.jdo.option.connectionURL=jdbc:hsqldb:hsql://localhost
lido.minPool=1
javax.jdo.option.ignoreCache=false
javax.jdo.option.multithreaded=false
javax.jdo.option.nontransactionalRead=true

Instatiating PersistenceManagerFactory

Listing JDO Vendor properties
VendorName=LIBeLIS
VersionNumber=1.2 build #022

Persisting BusinessPartner (number=1 name=Robin Roos address=Milton
Keynes)
Listing partners:
BusinessPartner (number=1 name=Robin Roos address=Milton Keynes)
Done.

LiDO C:\JDOSource\source\Ch02-SimpleExample\0201-SimpleExample>
```

AFTER CHAPTER 5

Most of the examples after Chapter 5, particularly the query examples in Chapter 8, require the domain as it is given for example 0506-GUI.

Thus when you run examples such as the DynamicQueryWindow (Example 0810) you should either be in the 0506 directory, or make the op.jdo persistence descriptor available in the 0810 directory.

OTHER IMPLEMENTATIONS

The next release of the JDOSource archive (due by 22 Nov 2002) will include scripts to support Kodo JDO, OpenFusion JDO and FastObjects j1.

For further information please contact: info@OgilviePartners.com